

One possible reason the college Wi-Fi is slow and frequently disconnects is that the Access Point (AP) is overloaded. If hundreds of students connect to the same AP at the same time, it may exceed its maximum client capacity. This causes network congestion, slower speeds, higher latency, and occasional disconnections. Another possibility is that the AP's coverage area is too large, resulting in weak signals for users who are far away.

How Upgrading One Device Could Help

Upgrading the Access Point to a newer Wi-Fi 6 or Wi-Fi 6E model with higher user capacity and better traffic management would improve performance. A modern AP can:

- Support many more connected devices simultaneously.

- Provide faster data transfer speeds.

- Reduce network congestion using advanced technologies such as OFDMA and MU-MIMO.

- Offer stronger coverage and more stable connections.

Example

Suppose 250 students are connected to a single old access point designed for only 50–60 users. During breaks or lectures, everyone uses the Wi-Fi at once, causing slow Internet and frequent disconnections. Replacing it with multiple high-capacity Wi-Fi 6 access points would distribute users across several APs, reducing congestion and providing faster, more reliable connectivity.